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Earth and Planetary Science Letters 175 (2000) 181–190

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Paleomagnetic directions in Late Precambrian glaciomarine sediments of the Mirbat Sandstone Formation, Oman

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Received 26 August 1999; received in revised form 29 November 1999; accepted 29 November 1999

Abstract

A paleomagnetic study has been carried out in the Late Precambrian glaciomarine Mirbat Sandstone Formation in the Sultanate of Oman. Limited availability and geographical distribution of outcrops restricted the sampling to six sites. Of these, three were in laminated, fine-grained clay- to siltstones and three in laminated carbonates. One of these sites was rejected. The remaining five sites, represented by a total of 21 samples, displayed a similar magnetic component during progressive thermal demagnetization between 0 and 500°C. The site mean pole calculated from this component is at 86.5°N 328.1°E ($\delta p = 11.2°$, $\delta m = 19.7°$). This low-temperature component is interpreted as the direction of the average recent geomagnetic field in Oman. Two sites (10 samples) revealed a consistent magnetic component above 500°C. The sample mean declination of 68.4° and inclination of 18.4° ($k = 46.4$, $\alpha_{95} = 7.2°$) are interpreted as the characteristic remanent magnetization. The Late Precambrian Arabian pole position computed from the samples at these two sites lies at 23.3°N 141.8°E ($\delta p = 3.9°$, $\delta m = 7.5°$). Rotated into West African coordinates, the resulting pole position is at 31.6°N 153.2°E, in good agreement with published poles from Gondwana of Late Precambrian age ($\sim$550 Ma). The resulting paleolatitude is about 9.4° indicating a low-latitude of deposition for the glaciomarine deposits of the Mirbat Sandstone Formation. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; Precambrian; pole positions; Arabian Shield; glaciation; glaciomarine sedimentation

1. Introduction and geological setting

Late Proterozoic and Early Paleozoic paleopole reconstructions, especially for Gondwana, are based on few data only, and are often of limited reliability [1,2]. Recent paleomagnetic studies (e.g. [3–5]) have improved our understanding of the distribution of the Gondwana continents in the Late Proterozoic and Early Paleozoic, even though the paleomagnetic database is still poor when compared to other cratons, e.g. Laurentia or Europe. Paleomagnetic data from the Arabian Shield, which became part of West-Gondwana after accretion took place around 640–600 Ma ago [6], are generally rare (e.g. [7]). With the co-operation of authorities in Oman we have carried out a limited paleomagnetic study that yields new results from an area where paleomagnetic data were previously non-existent.

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PII: S0012-821X(99)00307-6

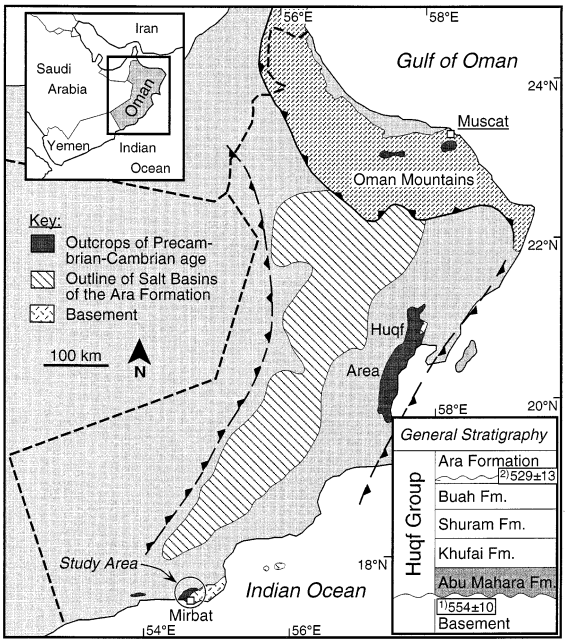


Fig. 1. Schematic structural map of the Sultanate of Oman, modified after [9]. The inset shows the general stratigraphy of the Late Precambrian to Cambrian Huqf Group. The investigated Mirbat Sandstone Formation is related to the Abu Mahara Formation [8]. The indicated radiometric ages [Ma] are data from 1)[13] and 2) unpublished data of Petroleum Development Oman.

Most Gondwana paleopoles are based on the study of igneous rocks, such as granitoids and volcanics. Only a few data have been derived from sediments since these are commonly deformed, metamorphosed, or otherwise inappropriate for paleomagnetic studies. In the southern part of the Sultanate of Oman (Fig. 1), unmetamorphosed siliciclastic deposits of Late Precambrian age are exposed and exceptionally well preserved. During a short sedimentological field campaign, an attempt has been made to determine the paleomagnetic directions in these rocks. The investigation was limited both in time and in geographical extent, but the results demonstrate the paleomagnetic potential of these fine-grained Late Precambrian sediments. Unfortunately, outcrops of lithologies that are suitable for paleomagnetic investigations (in particular, clay- to siltstones and limestones) are very limited in number, and geographical distribution, due to the variable sedimentary facies of the glaciomarine deposits in the investigated area. Moreover, the majority of the few outcrops consisting of mudstones were strongly weathered; their behavior when treated with a drilling device was extremely brittle. We sampled all available sites during our campaign. These were rather few, but to the best of our knowledge, it is not possible to increase the size of the database by acquiring further sites in the study area.

The study area is located in southern Oman, about 12–15 km northeast of Mirbat (Figs. 1 and 2). The investigated deposits comprise the Lower Member of the Mirbat Sandstone Formation, which is stratigraphically related to the Abu Mahara Formation at the base of the Huqf Group [8,9]. The Mirbat Sandstone Formation was deposited onto the tectonically stable Arabian craton. The sediments rest unconformably on Precambrian basement, which is mostly composed of gneisses and granodiorites [9,10]. The unmetamorphosed deposits provide well preserved primary sedimentary structures. The formation is cut by sand dikes, indicating a minor extensional phase, which, however, has no serious consequences for our paleomagnetic study. The post-depositional tilting of bedding is slight (about 8–10° towards northwest) and easily corrected.

The 400 m thick succession, cropping out below a steep cliff formed by Cretaceous and Tertiary carbonates (Jabal Samhan; Fig. 2), consists of well exposed glaciomarine sediments which were deposited at the southern border of the marine Abu Mahara Basin [9]. Glacial facies is indicated by a thick succession of massive diamictites containing glacially striated and faceted stones, laminated clay- and siltstones containing numerous dropstones, and the presence of typical features such as glacial striations on top of sandstone channels [11,12]. The diamicite units, several meters to tens of meters thick, are intercalated with numerous, channelized sandstones on the cm-to-m-scale. Fine-grained, laminated clay- to siltstones, as well as laminated carbonates (cm-to-dm-scale) are occasionally present; however they have limited lateral extent [11,12].

An age of 554 ± 10 Ma [13] has been derived by

$^{87}Rb/^{87}Sr$-dating of granodiorite dikes from the basement directly below the Abu Mahara Formation in the Huqf area (Fig. 1). Thus, deposition of the Mirbat Sandstone Formation did not start before about 560 Ma [14]. Pb–Pb dating carried out on carbonates from the base of the Ara Formation (top of the Huqf Group) revealed an age of 529 ± 16 Ma (unpublished data of Petroleum Development Oman). The possible age range of the deposits is therefore bracketed between 564 and 513 Ma. A mean value for the age of the Mirbat Sandstone Formation, which represents the base of the Huqf Group, is estimated to be about 560–540 Ma, indicating a Late Precambrian age [15,16].

2. Paleomagnetic methodology

2.1. Sampling

Attention was paid to the paleomagnetic suitability of the various lithologies. Fine-grained claystones, siltstones and carbonates that occur sporadically throughout the Lower Member of the Mirbat Sandstone Formation (Fig. 2) were sampled. However, due to the high lateral and vertical variability of the glaciomarine facies [11,12], as well as strongly disintegrating weathered surfaces, our sampling resulted in only six sites. The sampled lithologies were purple, laminated siltstone (site os-1); purple-gray, massive mudstone (site os-2); finely laminated, greenish-gray clay- to siltstone (site os-3); as well as laminated, light-gray to gray limestone (sites os-4, os-5 and os-6). Sampling was carried out with a portable water-cooled drilling machine. Each site is represented by 5–7 oriented cores, which were cut into 12–17 individual standard-sized one-inch paleomagnetic specimens in the laboratory. Site os-3 was sampled as two oriented blocks and drilled in the laboratory; it resulted in 4 cores, yielding seven individual specimens. Sample orientation was carried out with a Brunton compass. In total, we sampled six sites and obtained 24 samples which yielded 37 measurable specimens.

First, the natural remanent magnetization (NRM) of each specimen was measured. In a pilot study, 2–3 representative specimens from each site were thermally demagnetized at increments of 50°C between 100 and 500°C, thereafter every 30°C until 650°C, and in 10°C steps until 680°C. In addition, alternating field demagnetization (AF) was carried out on a set of 2–3 specimens from each site. The demagnetizing steps were at every 5 mT up to 30 mT, at 40 mT, and every 20 mT from 40 mT to 100 mT. The measurements were carried out with a three-axis 2G cryogenic magnetometer in a shielded room (noise level: $1–4 \times 10^{-6}$ A/m). The magnetic susceptibility was measured with a KLY-2 kappa-bridge following each heating step to monitor changes related to possible formation of magnetic minerals during heating. Based on the results of the pilot study of each site, all remaining samples were demagnetized thermally at the same steps as in the pilot study.

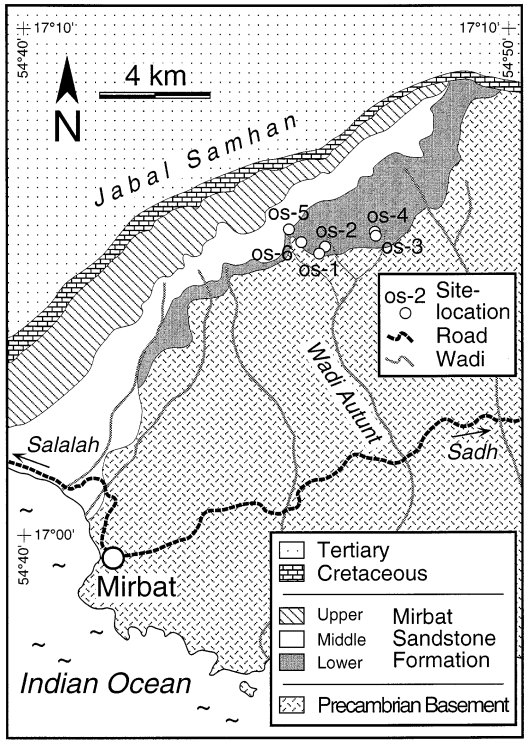


Fig. 2. Simplified geological map of the Mirbat area and location of sampled sites.

3. Results of the magnetic measurements

3.1. Pilot study

The specimens at all six sites displayed a well-defined magnetic direction up to about 450–500°C; this is shown as the low-temperature component in Fig. 3. The specimens at only two sites (os-2, os-6) possessed a stable magnetic component above 500°C, which is plotted as the high-temperature component in Fig. 3. It persisted until 660–680°C, and coincided with a general increase of the magnetic susceptibility indicating formation of new magnetic minerals (Fig. 3). This susceptibility increase was observed as a minor effect in the pilot specimens of sites os-2 but occurred strongly at os-6 above about 450°C. The measured directions, however, remained consistent and stable for the specimens of both sites (Fig. 3). At the four remaining sites there was no stable component above 500°C: either random directions were observed (Fig. 4), probably related to newly formed magnetic minerals (site os-4), or the signals were too weak to measure accurately (site os-5), or ambiguous and inconsistent directions ensued (os-1, os-3). Based on these detailed results, all specimens at sites os-2 and os-6 were demagnetized stepwise up to 680°C so as to define the high-temperature component in each specimen.

3.2. Interpretation of the low-temperature component

For samples that yielded more than one specimen the specimen results were averaged to give a

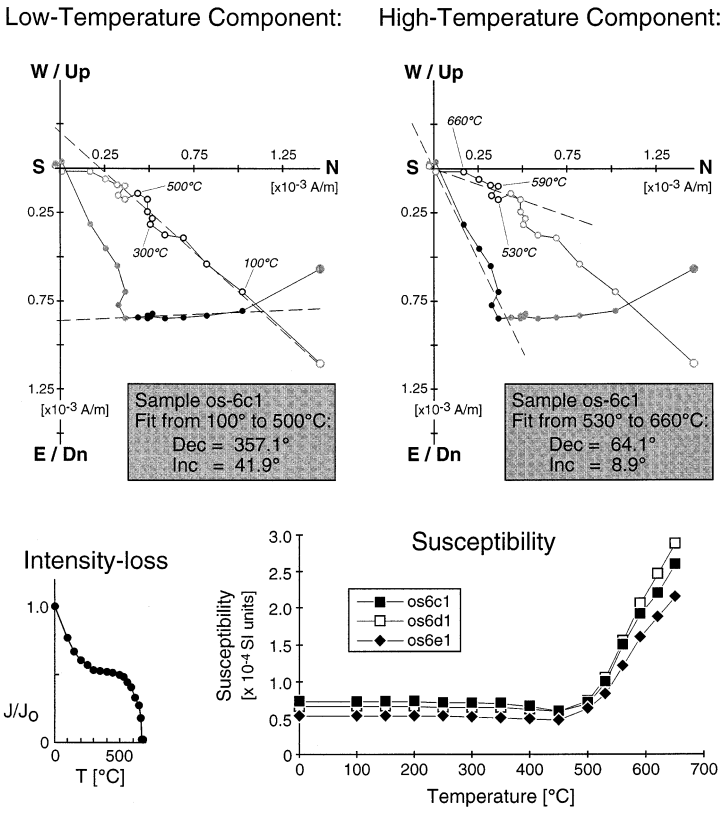


Fig. 3. Orthogonal vector demagnetization diagram [31], loss of remanent intensity, and change of magnetic susceptibility during heating of a specimen from site os-6. Open and filled circles represent projections of the magnetization on the (north-south) vertical and horizontal planes, respectively, for a given temperature step. Grayed symbols indicate data points that were not used to determine declination and inclination for each temperature interval.

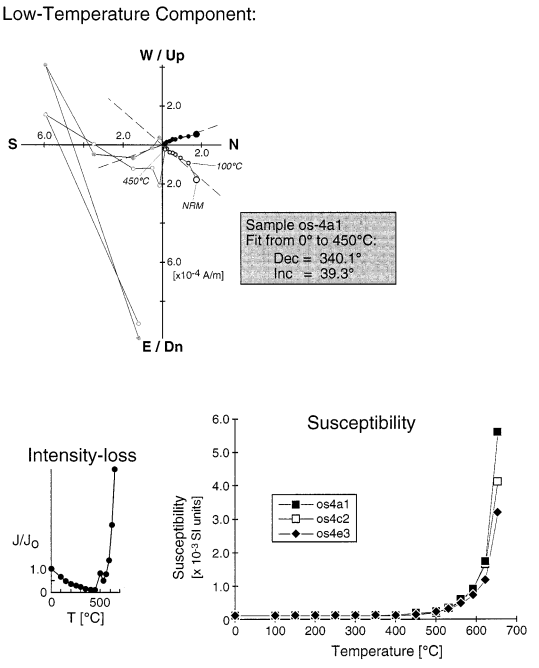


Fig. 4. Orthogonal vector demagnetization diagram [31], loss of remanent intensity, and change of magnetic susceptibility during heating of a specimen from site os-4. Symbols are explained in Fig. 3.

representative sample direction, and these were then combined to give site-mean directions. Overall mean directions, virtual geomagnetic poles (VGP) and statistical parameters [17,18] were calculated on the basis of both samples and sites (Table 1). Site os-5 consisted of only three specimens which gave divergent directions and unacceptable statistical properties (low k and very large α_{95}). The mean direction of the low-temperature component from the remaining five sites before bedding correction (Fig. 5, Table 2) has declination 356.3° and inclination 33.5° ($k = 16.4$, $\alpha_{95} = 17.3°$); the mean direction derived from the 21 samples (Fig. 6, Table 2) has declination 359.0° and inclination 35.8° ($k = 17.2$, $\alpha_{95} = 7.9°$). The corresponding paleomagnetic pole positions calculated from these overall mean directions are 86.5°N 328.1°E ($\delta p = 11.2°$, $\delta m = 19.7°$) and 88.1°N 26.6°E ($\delta p = 5.3°$, $\delta m = 9.2°$), respectively (Table 2); they are indistinguishable from the present-day axial dipole location. The VGP position was also calculated from each site mean direction. The mean of these five VGP lies at 86.8°N 341.6°E ($k = 17.0$, $\alpha_{95} = 17.3°$) in accordance with the data based on sample and site mean directions. The direction of the low-temperature component appears to be an overprint by the recent geomagnetic field. Moreover, comparing the statistical parameters for sample and site means before and after tilt-correction (Table 2) the data derived from the low-temperature component are slightly better grouped before tilt-correction which suggests post-tilting acquisi-

Table 1
Paleomagnetic results from the Mirbat Sandstone Formation, Oman: mean directions

Site	In situ data							Tilt-corrected data					
	N (n)	Dec (°)	Inc (°)	k	α_{95} (°)	VGP lat	VGP long	Dec (°)	Inc (°)	k	α_{95} (°)	VGP lat	VGP long
Low-temperature data													
os1	5 (7)	0.2	29.2	113.5	7.2	87.5°N	229.9°E	349.8	13.9	113.8	7.2	75.1°N	277.9°E
os2	4 (5)	3.8	19.5	102.4	9.1	81.1°N	209.2°E	1.7	12.7	101.6	9.2	78.2°N	226.2°E
os3	2 (5)	348.1	25.7	102.2	13.9	77.7°N	304.4°E	345.7	27.0	102.8	13.8	75.7°N	311.1°E
os4	4 (5)	334.5	37.7	42.3	14.3	65.8°N	335.9°E	325.4	36.0	33.6	16.1	57.3°N	333.4°E
os5	3 (3)	26.4	18.7	11.5	38.2	63.1°N	159.3°E	19.3	22.1	8.7	44.6	70.3°N	161.4°E
os6	6 (12)	18.7	51.4	37.7	11.1	68.1°N	101.2°E	4.5	54.1	37.8	11.0	73.0°N	67.4°E
High-temperature data													
os2	4 (5)	77.2	14.2	28.3	17.6	14.3°N	141.5°E	74.6	19.3	28.3	17.6	17.6°N	139.5°E
os6	6 (12)	65.4	7.5	92.8	7.0	24.6°N	148.7°E	64.4	17.6	93.7	7.0	27.0°N	143.4°E

Paleomagnetic mean directions and statistical parameters for the Mirbat Sandstone Formation, Oman, for in situ and tilt-corrected data. N and n give the number of samples and specimens, respectively. Dec and Inc are the declination and inclination of each component, respectively, k is the precision parameter and α_{95} the radius of confidence of the mean direction [18]. VGP lat and VGP long are latitude and longitude of calculated VGPs.

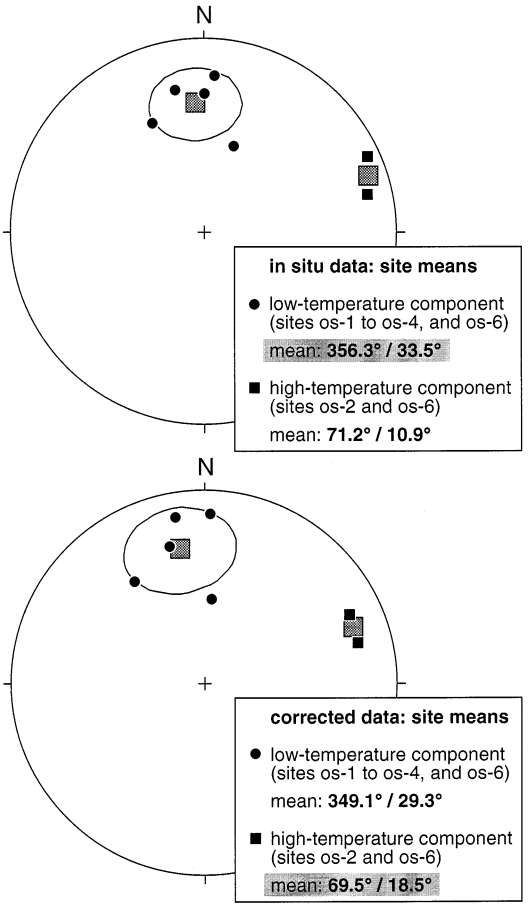


Fig. 5. Mean directions of site-mean magnetic components before (in situ) and after tilt-corrections. Statistical parameters are given in Tables 1 and 2.

tion of this component of remanent magnetization.

3.3. Interpretation of the high-temperature component

Combining the high-temperature components of sites os-2 (four samples, five specimens) and os-6 (six samples, 12 specimens) a tilt-corrected site mean direction with declination 69.5° and inclination 18.5° is obtained (Fig. 5, Tables 1 and 2). The pole position derived from this mean direction is at 22.3°N 141.4°E (Table 2). Because it is based on only two sites we cannot determine statistical parameters for this pole position. The mean direction calculated for the 10 individual samples at these sites has declination 68.4° and inclination 18.4° ($k = 46.4$, $\alpha_{95} = 7.2°$); the corresponding pole position (Table 2) lies at 23.3°N 141.8°E ($\delta p = 3.9°$, $\delta m = 7.5°$).

The grouping of the high-temperature component improves slightly after tilt-correction (Table 2). Although the improvement is not significant statistically, it does suggest that acquisition of remanent magnetization took place prior to tilting, probably during or shortly after deposition, and that this component represents the characteristic remanent magnetization. This interpretation is constrained by the field observation that site os-2 is slightly affected by soft-sediment deformation

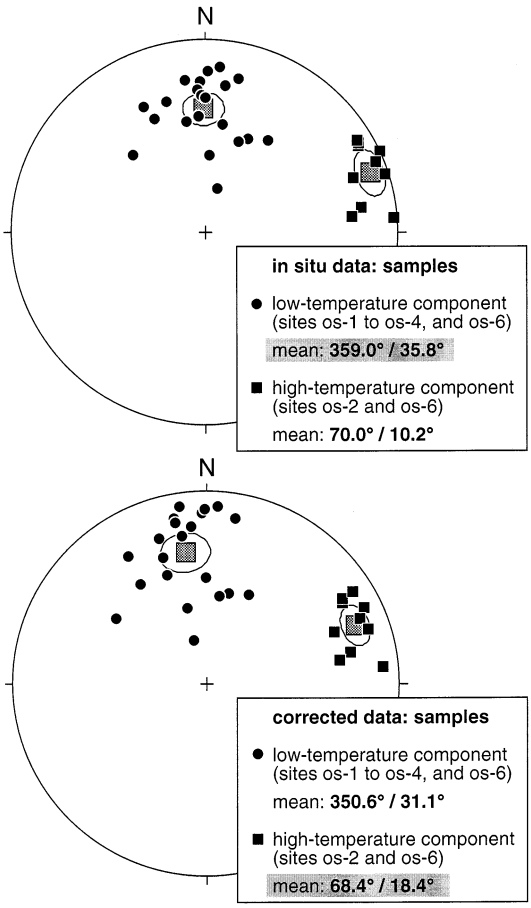


Fig. 6. Directions of magnetic components derived for individual samples before (in situ) and after tilt-corrections. Statistical parameters are given in Tables 1 and 2.

Table 2
Pole determination of Mirbat Sandstone Formation, Oman

	Database	N	Dec (°)	Inc (°)	k	α_{95} (°)	Lat	Long	δp (°)	δm (°)	Paleo-latitude
Low-temperature data											
In situ data	site mean directions	5	356.3	33.5	16.4	17.3	86.5°N	328.1°E	11.2	19.7	18.3°N
	Samples	21	359.0	35.8	17.2	7.9	88.1°N	26.6°E	5.3	9.2	19.8°N
Corrected data	site mean directions	5	349.1	29.3	11.5	20.8	79.3°N	313.1°E	12.7	23.0	15.7°N
	Samples	21	350.6	31.1	12.4	9.4	81.0°N	317.7°E	5.9	10.5	16.8°N
High-temperature data											
In situ data	site mean directions	2	71.2	10.9	–	–	19.5°N	145.0°E	8.4	16.5	5.5°N
	Samples	10	70.0	10.2	40.2	7.7	20.6°N	145.8°E	3.9	7.8	5.1°N
Corrected data	site mean directions	2	69.5	18.5	–	–	22.3°N	141.4°E	6.5	12.5	9.5°N
	Samples	10	68.4	18.4	46.4	7.2	23.3°N	141.8°E	3.9	7.5	9.4°N

Present-day site: lat 17°N, long 54.5°E
Site and sample mean directions and computed pole positions for the low-temperature and high-temperature magnetization components in the Mirbat Sandstone Formation, Oman. N is the number of sites or samples, Dec and Inc are the declination and inclination of mean directions, respectively, k is the precision parameter and α_{95} the radius of confidence of the mean direction [18]. Lat and Long are latitude and longitude of pole positions computed for the mean direction, δp and δm are the semi-minor and semi-major axes of the polar error ellipse.

(slump-like structures; see also [11,12]). We therefore assume that the high-temperature component represents the characteristic remanent magnetization. Only normal polarities were observed in our samples, so no reversal test of stability was possible.

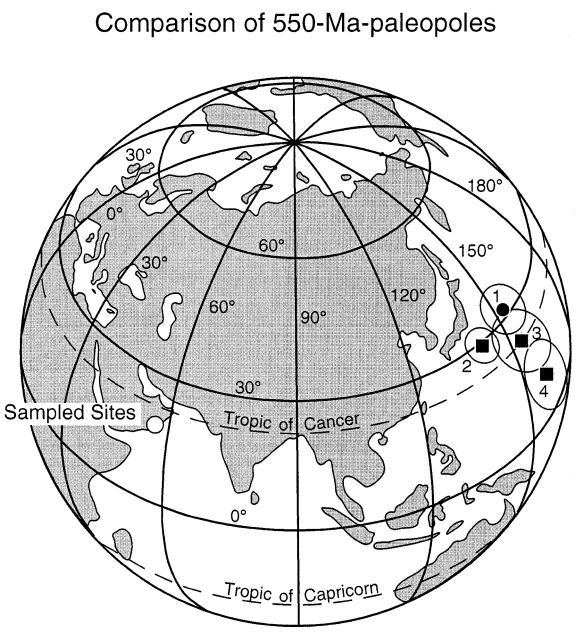
4. Discussion of results

The pole position calculated from the high-temperature component (23.3°N 141.8°E) must be corrected for younger movements of the Arabian peninsula. Rotation of the pole into West African coordinates was accomplished using the rotation parameters given by Lottes and Rowley [19]. These consist of a counterclockwise rotation of 7.12° about a rotation pole at 45.23°S 196.28°E for the transformation of Arabia into South African coordinates, and a clockwise rotation of 7.82° about a rotation pole at 9.34°N 5.7°E for transformation of South African into West African coordinates. These transformations give a Late Precambrian pole position of 31.6°N 153.2°E for the Arabian peninsula (Fig. 7). A recently published collection of reliable paleomagnetic poles from Gondwanan blocks contains some (south) poles for the interval 550–510 Ma (see table 3 and figure 14b in [3]). Transformed into north poles, the published poles at $\sim$550 Ma agree well with our data from the Mirbat Sandstone Formation (Fig. 7), whereas the younger poles of age 530–510 Ma do not fit well. In fact our calculated pole does not agree with any known pole positions of younger age, thus excluding the likelihood that our characteristic remanent magnetization might be a later remagnetization.

Seven criteria have been established for assessing the reliability of paleomagnetic poles [1,2]. They include age control of the magnetization and statistical requirements.

Age control can be a major problem in old rocks, especially in Proterozoic strata. Radiometric dating has suggested an age of 560–540 Ma for the Mirbat Sandstone Formation. However, the quoted ages are derived from the Huqf area and correlated to the Mirbat area, although with great confidence.

The statistical requirements are met only marginally, largely because of the limited sampling



Pole		Age [Ma]	Lat.	Long.	α_{95}
1	Oman	~550	32°N	153°E	7°
2	Congo	547 ± 4	29°N	139°E	5°
3	India	~550	23°N	153°E	11°
4	Australia	550 ± 15	12°N	163°E	13°

Fig. 7. Comparison of published paleopole data ([3], south poles transformed to north poles) with the mean pole derived from the Mirbat Sandstone Formation, Oman (rotated into West African coordinates).

that was possible. The number of data should exceed 24, the precision parameter k should be greater than 10, and the circle of confidence should be less than 16° [2]. The number of data in our study is too few for both the low-temperature component (21 samples, five sites) and the high-temperature component (10 samples, two sites). However, the precision parameters for the high- and low-temperature components are greater than 10 at all five usable sites and the angles of confidence are less than 16° at most of these. Clearly, having only two sites for the evaluation of the high-temperature component is a weakness of our study. However, the overall averages based on samples meet the statistical requirements except for the number of data. The results add a new datapoint for an area that has not until now been represented.

Our results agree well with existing paleomagnetic data for the reconstruction of Gondwana around 550 Ma ago. They indicate that Arabia occupied more or less its present-day position with respect to Africa at that time. Several authors of reconstructions of Gondwana (e.g. [4,20]) have assumed that, apart from the effects of the opening of the Red Sea, the positions of Arabia and Africa were adjacent. Our study confirms this inference.

The results add to the body of literature in which paleomagnetically determined low latitudes have been determined for Late Proterozoic rocks of undoubted glacial origin (e.g. [21–23]). In order to explain the paradoxical scenario of widespread glaciation in equatorial latitudes [24] several different mechanisms have been suggested. The 'snowball Earth' model postulates that the oceans froze worldwide [25,26]. This could have happened in different ways.

Paleomagnetic results show that from ~750 to ~550 Ma ago the continents were grouped to form a single supercontinent, Rodinia. Fast plate motions accompanying its break up would have caused rapid burial of organic material on continental margins, which in turn could have resulted in depletion of atmospheric CO_2 and a global lowering of temperatures so that global freezing occurred [26]. An alternative model supposes that the long-lived existence of a single continent might have modified the Earth's oblateness and thereby its obliquity, so that the latter exceeded ~54° [27,28]. The change in insulation would have caused glaciation in equatorial latitudes.

An alternative explanation for the equatorial glaciations is that they are due to 'inertial-interchange true polar wander' [29] in which the Earth's major and intermediate principal axes of inertia are exchanged. This type of event could be triggered episodically whenever anomalously fast plate motions result in an unstable configuration of the inertia axes. To restore equilibrium the major axis would shift rapidly through 90°, resulting in rotation of the lithosphere and mantle relative to the spin axis.

These models intrinsically assume the correctness of the paleomagnetic interpretation that measured shallow inclinations of magnetization

imply a low-latitude origin. This in turn depends on the validity of the geomagnetic axial dipole (GAD) hypothesis, whereby the long-term averaged paleomagnetic field as recorded in rocks corresponds to that of a dipole oriented along the Earth's rotation axis. This hypothesis holds well for Cenozoic and Mesozoic rocks, although a minor quadrupole component may also be present. In a recent evaluation of the entire paleomagnetic database Kent and Smethurst [30] found that Paleozoic and Precambrian data are characterized by shallower inclinations than expected for a GAD field. The effects could be modelled by adding substantial axial octupole and minor quadrupole components to the axial dipole. This implies that the GAD hypothesis may not be applicable for the Precambrian, casting doubt on paleomagnetic interpretations and reconstructions for that time. Kent and Smethurst [30] point out that their analytical model assumes that the long-term average motions of the continents are random. If they congregate for long intervals, as in a preferred distribution along the equator, the statistical analysis would itself break down.

6. Conclusions

Our paleomagnetic study of glaciomarine deposits from the Sultanate of Oman was restricted in scope, yet the few results are likely to be unique for this area. It is virtually impossible to collect further sites from this study area. The number of outcrops with lithologies suitable for paleomagnetic studies is very limited. Moreover, the lithology changes rapidly both vertically and laterally so that most sampled sites could be traced laterally for only a few meters.

The samples contained two magnetic components. The low-temperature component is interpreted to be a secondary magnetization related to the recent or modern geomagnetic field. The high-temperature component is interpreted as the characteristic remanent magnetization of probable primary origin. It yields a Late Precambrian pole position for the Arabian peninsula at 31.6°N 153.2°E. The compatibility of this with other Gondwana poles may be regarded as evidence that the relative position of Arabia with respect to Africa 550 Ma ago was similar to the present configuration. The observed inclination of the characteristic remanent magnetization implies deposition of the glaciomarine Mirbat Sandstone Formation at a very low latitude ($\sim$9.4°). This result is compatible with other observations of shallow inclinations of magnetization in Late Precambrian glaciogenic deposits.

Acknowledgements

We are grateful to Mohammed Kassim and Dr. Hilal bin Mohammed Al-Azri (Directorate General of Minerals, Ministry of Industry and Commerce, Oman) for support of this project and to J. Amthor (Petroleum Development Oman) for permission to use the unpublished radiometric age date of the Ara Formation. We thank F. Heller and A.M. Hirt for critically reading the manuscript and B. Geyer for technical support. The first author also thanks P. Strunck and S. Schröder (Bern) for stimulating discussions. Critical reviews by J. Meert and C. Mac Niocaill improved the manuscript and are gratefully acknowledged. Contribution 1094, Institute of Geophysics, ETH Zürich.*[RV]*

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